

Niveau Facile — Corrigé

Corrigé 1

$\int x e^x dx$. **Choix :** $U = x$, $V' = e^x$, donc $U' = 1$, $V = e^x$.

$$\int x e^x dx = x e^x - \int e^x \cdot 1 dx = x e^x - e^x + C = \boxed{(x-1)e^x + C}.$$

Vérif. $[(x-1)e^x]' = e^x + (x-1)e^x = x e^x$. ✓

Corrigé 2

$\int x \cos x dx$. **Choix :** $U = x$, $V' = \cos x$, donc $U' = 1$, $V = \sin x$.

$$\int x \cos x dx = x \sin x - \int \sin x \cdot 1 dx = x \sin x - (-\cos x) + C = \boxed{x \sin x + \cos x + C}.$$

Vérif. $[x \sin x + \cos x]' = \sin x + x \cos x - \sin x = x \cos x$. ✓

Corrigé 3

$\int x \sin x dx$. **Choix :** $U = x$, $V' = \sin x$, donc $U' = 1$, $V = -\cos x$.

$$\int x \sin x dx = x(-\cos x) - \int (-\cos x) \cdot 1 dx = -x \cos x + \int \cos x dx = \boxed{-x \cos x + \sin x + C}.$$

Vérif. $[-x \cos x + \sin x]' = -\cos x + x \sin x + \cos x = x \sin x$. ✓

Corrigé 4

$\int x e^{-x} dx$. **Choix :** $U = x$, $V' = e^{-x}$, donc $U' = 1$, $V = -e^{-x}$.

$$\int x e^{-x} dx = x(-e^{-x}) - \int (-e^{-x}) dx = -x e^{-x} + \int e^{-x} dx = -x e^{-x} - e^{-x} + C = \boxed{-(x+1)e^{-x} + C}.$$

Vérif. $[-(x+1)e^{-x}]' = -e^{-x} - (x+1)(-e^{-x}) = -e^{-x} + (x+1)e^{-x} = x e^{-x}$. ✓

Corrigé 5

$\int (x+2) e^x dx$. **Choix :** $U = x+2$, $V' = e^x$, donc $U' = 1$, $V = e^x$.

$$\int (x+2) e^x dx = (x+2) e^x - \int e^x dx = (x+2) e^x - e^x + C = \boxed{(x+1) e^x + C}.$$

Vérif. $[(x+1)e^x]' = e^x + (x+1)e^x = (x+2)e^x$. ✓